

MONKES
(MONOENERGETIC KINETIC EQUATION SOLVER)
MANUAL
(DRAFT)

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1 Installation and required libraries

In this section it is sketched how to install MONKES.

1.1 Fortran compiler

MONKES is written in Fortran language and thus requires having an installed Fortran compiler. This compiler must be compatible with LAPACK and NetCDF libraries. One possible compiler is the Intel Fortran compiler.

1.2 NetCDF library

In order to read BOOZER_XFORM output files, MONKES needs to use the NetCDF library. An example of the minimal required libraries for running MONKES in a UNIX machine is displayed below. Modules 1) and 2) are the NetCDF library compatible with the Fortran compiler. Module 3) is the Fortran compiler version, in this case intel17/17.0.8.

Currently Loaded Modules:

- 1) netcdf-fortran-4.4.4-intel-17.0.8-4me7upi
- 2) netcdf-4.6.1-intel-17.0.8-i5cn5xw
- 3) intel17/17.0.8

1.3 LAPACK library

For using LAPACK library there are two options. One can use the static or the dynamic version of the library. In listing 1, the dynamic case is illustrated.

```
# *** Compiler selection
f90comp=gfortran
NETCDF_DIR=/usr/include/

FFLAGS=-I$(NETCDF_DIR) -mcmodel=large -O2 -freal-4-real-8 -ffree-line-length-512
LFLAGS=-llapack -lnetcdf -lnetcdff -lblas
```

Listing 1: Makefile

The variable LFLAGS includes the linking of LAPACK libraries. Specifically, those flags which include the variable \$MKL_HOME (variable containing the location of the dynamic library) set the instructions for where to find the library. The flag -mkl=parallel allows for LAPACK multithreading functionalities. In listing 2, it is shown how the monkes executable is generated.

```
main_monkes.x: $(objects)
    $(f90comp) -o main_monkes.x $(objects) ${LFLAGS} ${FFLAGS}
```

Listing 2: Makefile

2 How to use MONKES

2.1 Magnetic configuration input

MONKES can read the magnetic configuration of a particular flux surface from different types of files.

2.1.1 From BOOZER_XFORM output file

MONKES can read the magnetic configuration of an output (binary) file from BOOZER_XFORM. The file **must** be named boozmn.nc. It is advisable to make a soft link instead of renaming the original BOOZER_XFORM output file.

In an output file from BOOZER_XFORM there is information about the equilibrium of many flux surfaces. If not specified, MONKES will perform its calculations at the flux surface labelled by $s = \psi/\psi_{\text{lcs}} = 0.25$ where $2\pi\psi$ is the toroidal flux and ψ_{lcs} the label of the last closed flux surface. A different flux surface can be specified in the input file monkes_input.surface. An example on how to fill the input file monkes_input.surface is given in listing 3.

```
&surface
s=0.2
/
```

Listing 3: Example of `monkes_input.surface` for specifying the surface $s = \psi/\psi_{\text{lcs}} = 0.2$

2.1.2 From DKES input file

The magnetic configuration can also be specified using a DKES input file. This input file contains the information about a single flux surface and therefore s does not need to be (explicitly) specified. For running MONKES using a DKES input file, it **must** be renamed as `ddkes2.data`.

2.2 Monoenergetic database input

In addition to the magnetic configuration (see section 2.1), in order to run MONKES for calculating the monoenergetic coefficients $\hat{D}_{ij}$, it is necessary to specify 5 additional parameters. The required parameters are listed in table 1.

These parameters have to be specified in the input file `monkes_input.parameters`. For each combination of $(N_\theta, N_\zeta, N_\xi, \hat{\nu}, N_\xi)$ MONKES will compute the monoenergetic coefficients $\hat{D}_{ij}$. An example on how to fill this input file is given in listing 4. Note that in the example shown in listing 4 there is a single value for `N_theta`, `N_zeta` and `N_xi` while there are 10 values `nu` and 2 values of `E_r`. This means that MONKES will perform a scan in $\hat{\nu}$ and $\hat{E}_r$ for fixed resolution $(N_\theta, N_\zeta, N_\xi) = (30, 60, 160)$. Similarly, one could be interested in doing scans in the different resolutions to check how the monoenergetic coefficients $\hat{D}_{ij}$ converge. This scan is also possible by introducing several values of, say N_ζ , separated by commas.

```
&parameters
N_theta = 30
N_zeta = 60
N_xi = 160
nu =
1.000000e-05,
3.000000e-05,
1.000000e-04,
3.000000e-04,
1.000000e-03,
3.000000e-03,
1.000000e-02,
3.000000e-02,
1.000000e-01,
3.000000e-01
E_r = 0.000000e+00, 3e-4
/
```

Listing 4: Example of `monkes_input.parameters` for calculating a monoenergetic database

2.3 Monoenergetic database output

The monoenergetic coefficients $\hat{D}_{ij}$ are written in an output file called `monkes_Monoenergetic_Database.dat`. Instead of writing directly $\hat{D}_{ij}$ which are in meters, in the output file `monkes_Monoenergetic_Database.dat`,

Parameter	Symbol in MONKES paper [1]	Name in MONKES input file	Units
Poloidal resolution	N_θ	<code>N_theta</code>	N/A
Toroidal resolution	N_ζ	<code>N_zeta</code>	N/A
No. Legendre modes	N_ξ	<code>N_xi</code>	N/A
Collisionality	$\hat{\nu}$	<code>nu</code>	1/m
Radial electric field	$\hat{E}_r$	<code>E_r</code>	V · m/s

Table 1: Parameters to be specified in the file `monkes_input.parameters` for calculating the monoenergetic coefficients.

the quantities written are $K_{ij}\hat{D}_{ij}$. The monoenergetic coefficients are defined in section 2 of reference [1]. The quantities K_{ij} , as defined in Appendix E of [1], are: $K_{11} = (\mathrm{d}\psi/\mathrm{d}r)^{-2}$, $K_{13} = K_{31} = (\mathrm{d}\psi/\mathrm{d}r)^{-1}$ and $K_{33} = 1$.

nu/v [m ⁻¹]		E _r /v [V s /m ²]		N _{theta}	N _{zeta}	N _{xi}	
D ₁₁		D ₃₁		D ₁₃	D ₃₃	D _{33_Spitzer}	Wall-clock
time [s]	CPU time [s]						
0.1000000000000000E-04	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.8083370894646827E-01	0.3558097167954141E+00			-0.3557974181173633E+00	0.3781812078503804E+05	0.6628557164949892E+05	
0.4385000000000000E+02	0.5232387820000000E+03						
0.3000000000000000E-04	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.2810661936903304E-01	0.3256727047296022E+00			-0.3256699118937129E+00	0.1291631323161333E+05	0.2209519054983298E+05	
0.4500700000000000E+02	0.5381165510000001E+03						
0.1000000000000000E-03	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.9699155660889601E-02	0.2329025257396027E+00			-0.2329024600913502E+00	0.4016302448315862E+04	0.6628557164949895E+04	
0.4472800000000000E+02	0.5347238680000000E+03						
0.3000000000000000E-03	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.4279607172778349E-02	0.8845086936896115E-01			-0.8845086932979547E-01	0.1408020116218274E+04	0.2209519054983297E+04	
0.4602000000000000E+02	0.5501236660000002E+03						
0.1000000000000000E-02	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.2447678833565718E-02	-0.5447447639450777E-02			0.5447447639461694E-02	0.4599905179277028E+03	0.6628557164949892E+03	
0.4708800000000000E+02	0.5628036520000001E+03						
0.3000000000000000E-02	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.2456221430843609E-02	-0.3895792255643186E-01			0.3895792255645109E-01	0.1702769606149794E+03	0.2209519054983300E+03	
0.4869800000000000E+02	0.5820463040000000E+03						
0.1000000000000000E-01	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.3404582820706755E-02	-0.3962755023427194E-01			0.3962755023427266E-01	0.5800868190595923E+02	0.6628557164949895E+02	
0.4675700000000000E+02	0.5583137999999999E+03						
0.3000000000000000E-01	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.4750847233951999E-02	-0.2123155106854335E-01			0.2123155106854411E-01	0.2099408256854233E+02	0.2209519054983298E+02	
0.4713000000000000E+02	0.5633311179999996E+03						
0.1000000000000000E+00	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.8203825856789445E-02	-0.6314388928966534E-02			0.6314388928967790E-02	0.6540004603642041E+01	0.6628557164949894E+01	
0.4763800000000000E+02	0.5692957600000000E+03						
0.3000000000000000E+00	0.0000000000000000E+00			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.1908565050702955E-01	-0.1079732003337501E-02			0.1079732003338740E-02	0.2204175406018691E+01	0.2209519054983298E+01	
0.4679100000000000E+02	0.5591580839999997E+03						
0.1000000000000000E-04	0.3000000000000000E-03			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.1400431461308468E-01	0.2321125616633982E+00			-0.2321089946168132E+00	0.3785637549091421E+05	0.6628557164949892E+05	
0.4883100000000000E+02	0.5835083770000001E+03						
0.3000000000000000E-04	0.3000000000000000E-03			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.1560611355971947E-01	0.2809610046367516E+00			-0.2809607056510192E+00	0.1293144634657525E+05	0.2209519054983298E+05	
0.4701800000000000E+02	0.5618436950000005E+03						
0.1000000000000000E-03	0.3000000000000000E-03			0.3100000000000000E+02	0.6100000000000000E+02	0.1600000000000000E+03	
0.8697565729005768E-02	0.2156076558873087E+00			-0.215			

Listing 5: Example of `monkes_Monoenergetic_Database.dat` for calculating a monoenergetic database

2.4 Example of running MONKES for Wendelstein 7-X EIM geometry

In the folder location `examples/W7X-EIM/` there is an example for running MONKES in Wendelstein 7-X (W7-X) geometry. For both cases, the `monkes_input.parameters` file used is the one shown in listing 4. That is, a scan of 10 points in $\hat{\nu}$ for 2 different values of the radial electric field $\hat{E}_r$.

2.4.1 Running from DKES input

An example in which the magnetic configuration is specified via a DKES input file can be found at `examples/W7X-EIM/dkes_input/`. Once that MONKES has been correctly compiled, this example can be run by simply typing

```
./../../../../bin/main_monkes.x
```

in the command prompt or, if running in a remote platform, by executing the batch file from within this location.

The monoenergetic coefficients, are written in the `monkes_Monoenergetic_Database.dat` file. Specifically, the results of this run correspond to the output file example of listing 5. In figure 1, the main results of the simulation are plotted, for both values of the electric field $\hat{E}_r$, against the collisionality $\hat{\nu}$.

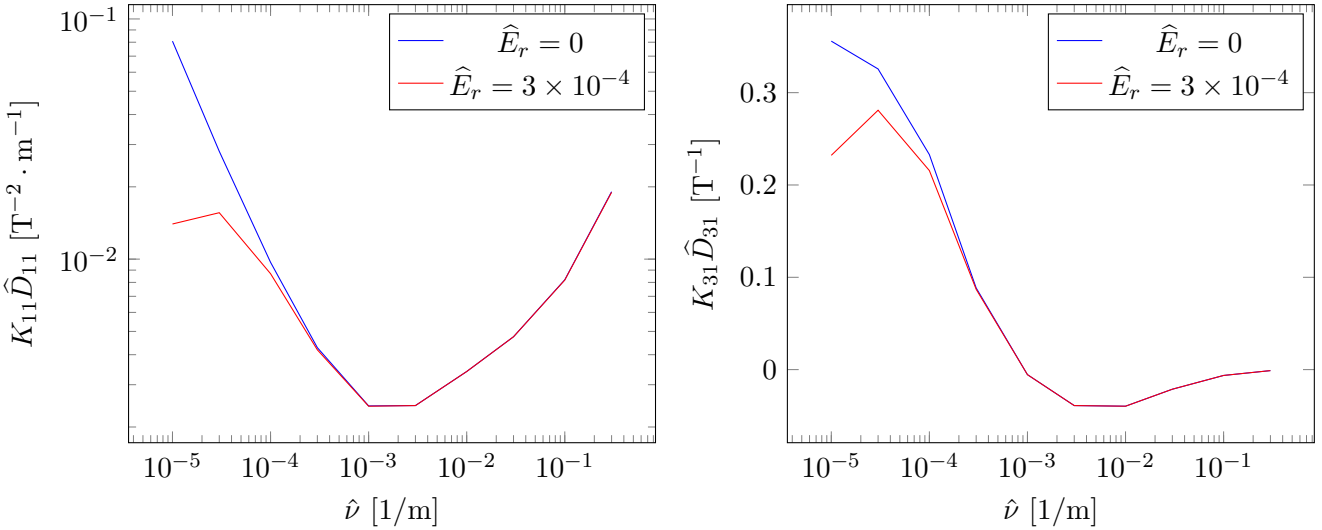


Figure 1: Radial transport $K_{11}\hat{D}_{11}$ (left) and bootstrap current coefficient $K_{31}\hat{D}_{31}$ (right) obtained from running the example using a DKES input file.

2.4.2 Running from BOOZER_XFORM output

An example in which the magnetic configuration is specified via a BOOZER_XFORM output file can be found at `examples/W7X-EIM/boozer_xform_input/`. In this folder, the file `boozmn_wout_w7x_std_v_0.0.txt.nc` contains the magnetic configuration. As mentioned in section 2.1 MONKES expects any BOOZER_XFORM output file to be named `boozmn.nc`. Thus, before running MONKES, the file `boozmn_wout_w7x_std_v_0.0.txt.nc` should be renamed to `boozmn.nc`. The surface $\psi/\psi_{\text{lcs}} = 0.2$ is specified in the `monkes_input.surface` file as shown in listing 3. This surface also corresponds to the one specified in the DKES input file from section 2.4.2.

Now, provided that MONKES has been correctly compiled, this example can be run by simply typing

```
./../../../../bin/main_monkes.x
```

in the command prompt or, if running in a remote platform, by executing the batch file from within this location.

In figure 2, the main results of the simulation (written in the `monkes_Monoenergetic_Database.dat` file) are plotted.

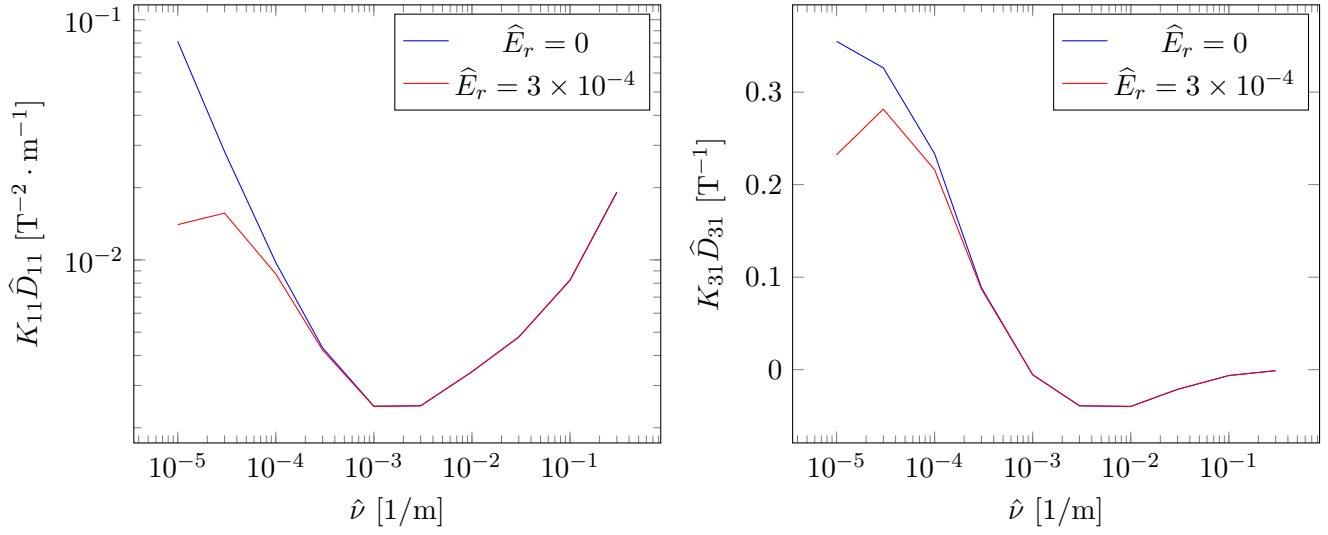


Figure 2: Radial transport $K_{11}\hat{D}_{11}$ (left) and bootstrap current coefficient $K_{31}\hat{D}_{31}$ (right) obtained from running the example using a BOOZER_XFORM output file.

References

- [1] F.J. Escoto, J.L. Velasco, I. Calvo, M. Landreman, and F.I. Parra. MONKES: a fast neoclassical code for the evaluation of monoenergetic transport coefficients in stellarator plasmas. *Nuclear Fusion*, 64(7):076030, 2024. [doi:10.1088/1741-4326/ad3fc9](https://doi.org/10.1088/1741-4326/ad3fc9).